

Further Integration Techniques



Integration by substitution

- 1 Evaluate $\int 2x(x^2 + 1)^4 dx$ using the substitution $u = x^2 + 1$.
 - 2 Evaluate $\int_0^{\pi/2} \sin^3 x \cos x dx$ using the substitution $u = \sin x$.
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Integration by parts

- 3 Evaluate $\int x e^x dx$ using integration by parts.
 - 4 Evaluate $\int x \ln x dx$ using integration by parts.
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Repeated integration by parts

- 5 Evaluate $\int x^2 e^x dx$, applying integration by parts twice.
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Integration using trigonometric identities

- 6 Evaluate $\int \cos^2 x dx$ using the identity $\cos^2 x = (1 + \cos 2x)/2$.
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Integration using partial fractions

- 7 Evaluate $\int 1/(x^2 - 1) dx$ by first expressing the integrand in partial fractions.
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Evaluating a definite integral

- 8 Evaluate $\int_1^e \ln x/x dx$, using the substitution $u = \ln x$.
 - 9 Evaluate $\int_0^{\pi} x \sin x dx$ using integration by parts.
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Integration involving inverse trigonometric results

- 10 Evaluate $\int 1/(1 + x^2) dx$, and use it to evaluate $\int_0^1 1/(1 + x^2) dx$.

Integration by substitution with trigonometric functions

- 11 Evaluate $\int \sin^3 x \, dx$, using the identity $\sin^2 x = 1 - \cos^2 x$ and the substitution $u = \cos x$.
- 12 Evaluate $\int \tan x \, dx$, using the substitution $u = \cos x$.

Integration by parts with a logarithmic integrand

- 13 Evaluate $\int_1^e \ln x \, dx$ using integration by parts.

Integration using a Weierstrass-style substitution

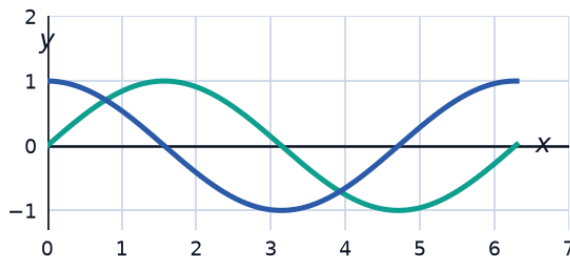
- 14 Evaluate $\int 1/(5 + 4\cos x) \, dx$ using the substitution $t = \tan(x/2)$, given that this substitution transforms $\cos x = (1 - t^2)/(1 + t^2)$ and $dx = 2/(1 + t^2)dt$, leading to the simplified integrand $2/(9 + t^2)$.

Integration by parts with a trigonometric-exponential product

- 15 Evaluate $\int e^x \cos x \, dx$, applying integration by parts twice and solving the resulting equation for the original integral.

Finding the area between two curves using integration

- 16 Find the area of the region enclosed between $y = \sin x$ and $y = \cos x$ for $\pi/4 \leq x \leq 5\pi/4$ (the interval on which $\sin x \geq \cos x$).



- 17 Evaluate $\int x^2 \ln x \, dx$ using integration by parts.
- 18 A particle's velocity is given by $v(t) = t \cos t$ metres per second. Find the particle's displacement over the interval $0 \leq t \leq \pi$, using integration by parts.

- 19 This question has three parts. (a) Use the substitution $u = x^2 + 4$ to find $\int x/(x^2 + 4) dx$. (b) Use integration by parts to find $\int \arctan x dx$. (c) Hence, using the result of part (b) and the standard result $\int 1/(1 + x^2) dx = \arctan x + C$, evaluate $\int_0^1 \arctan x dx$, giving an exact answer in terms of π and $\ln$.
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Reduction formulas

- 20 Given the reduction formula $I_n = \int x^n e^x dx = x^n e^x - n I_{n-1}$ (obtained via integration by parts), and that $I_0 = e^x + C$, use it to find $I_2 = \int x^2 e^x dx$.
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Integration by substitution with a linear inner function

- 21 Evaluate $\int (3x + 2)^5 dx$ using the substitution $u = 3x + 2$.
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Integration using the sine double-angle identity

- 22 Evaluate $\int \sin^2 x dx$ using the identity $\sin^2 x = (1 - \cos 2x)/2$.
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Volume of revolution using integration

- 23 The region bounded by $y = \sqrt{x}$, the x -axis, and $x = 4$ is rotated 360° about the x -axis. Find the volume of the resulting solid, using $V = \pi \int_0^4 y^2 dx$.
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Substitution with a shifting definite integral bound

- 24 Evaluate $\int_0^1 x \sqrt{x^2 + 1} dx$ using the substitution $u = x^2 + 1$.
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Choosing an appropriate integration technique

- 25 Which integration technique is most appropriate for evaluating $\int x^3 \sin x \, dx$?
- a. Integration by parts, applied three times (reducing the power of x each time)
 - b. Substitution $u = \sin x$
 - c. Partial fractions
 - d. The double-angle identity for $\sin$